Benchmark comparison tables — acquisition × surrogate model

Legend. Each cell is the final metric (mean ± sd over seeds). **Bold** marks the best model for that acquisition; green marks the single best (acquisition, model) combination for that problem. Lower is better for both metrics.

Table 1: Final regret (value − f^*) for every acquisition × surrogate model ($n_{\text{rep}} = 3$, mean ± sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	**0.00641 ± 0.0063**	0.0911 ± 0.057	**0.00413 ± 0.015**	0.0322 ± 0.12
	Heteroscedastic LVGP	0.0302 ± 0.035	0.0913 ± 0.057	0.00858 ± 0.021	0.0326 ± 0.12
	Separate GP	0.624 ± 1.2	0.649 ± 1.1	0.0912 ± 0.1	0.172 ± 0.33
	Categorical kernel	0.196 ± 0.71	**0.0516 ± 0.088**	0.00441 ± 0.015	**0.0168 ± 0.087**
PI	Standard LVGP	**0.0127 ± 0.015**	**0.0697 ± 0.057**	**0.000736 ± 0.0011**	**0.000135 ± 0.00025**
	Heteroscedastic LVGP	0.547 ± 1.2	0.289 ± 0.91	0.0395 ± 0.072	0.0345 ± 0.12
	Separate GP	0.12 ± 0.55	0.179 ± 0.42	0.026 ± 0.048	0.06 ± 0.18
	Categorical kernel	0.0145 ± 0.012	0.0736 ± 0.088	0.000795 ± 0.0019	0.00107 ± 0.0013
EI	Standard LVGP	**0.0123 ± 0.012**	**0.0532 ± 0.054**	**$3.11 \times 10^{-7} \pm 3.7 \times 10^{-6}$**	**$7.92 \times 10^{-6} \pm 3.3 \times 10^{-5}$**
	Heteroscedastic LVGP	0.0212 ± 0.024	0.092 ± 0.049	0.00697 ± 0.019	0.0369 ± 0.12
	Separate GP	0.221 ± 0.76	0.419 ± 0.71	0.0292 ± 0.049	0.0326 ± 0.12
	Categorical kernel	0.0132 ± 0.014	0.0898 ± 0.092	0.000435 ± 0.00059	0.000441 ± 0.00073
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0166 ± 0.016	0.0862 ± 0.054	0.00464 ± 0.016	**0.000791 ± 0.0013**
	Separate GP	0.217 ± 0.77	0.0966 ± 0.12	0.0271 ± 0.049	0.0462 ± 0.18
	Categorical kernel	**0.0122 ± 0.012**	**0.0842 ± 0.088**	**0.00088 ± 0.0015**	0.000906 ± 0.0011
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.024 ± 0.025	**0.078 ± 0.085**	**0.0137 ± 0.024**	**0.000224 ± 0.00067**
	Separate GP	0.027 ± 0.027	0.19 ± 0.39	0.0467 ± 0.061	0.0341 ± 0.12
	Categorical kernel	**0.0191 ± 0.024**	0.131 ± 0.41	0.0267 ± 0.029	0.00284 ± 0.0053
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	**0.0264 ± 0.024**	**0.00249 ± 0.0073**	0.00853 ± 0.021	0.017 ± 0.087
	Separate GP	0.723 ± 1.3	0.377 ± 0.91	0.1 ± 0.12	0.188 ± 0.33
	Categorical kernel	0.11 ± 0.55	0.0269 ± 0.081	**0.00463 ± 0.015**	**0.000752 ± 0.002**

Table 2: Final regret (value $-$ f^*) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	$\mathbf{0.00408 \pm 0.0047}$	$\mathbf{0.00124 \pm 0.0059}$	$\mathbf{0.00413 \pm 0.015}$	0.0322 ± 0.12
	Heteroscedastic LVGP	0.0188 ± 0.021	0.0993 ± 0.063	0.0105 ± 0.023	0.0323 ± 0.12
	Separate GP	0.726 ± 1.3	0.529 ± 0.88	0.0769 ± 0.094	0.2 ± 0.33
	Categorical kernel	0.294 ± 0.88	0.0941 ± 0.12	0.0183 ± 0.06	$\mathbf{0.0185 \pm 0.098}$
PI	Standard LVGP	0.0131 ± 0.014	$\mathbf{0.00462 \pm 0.023}$	$\mathbf{0.0009 \pm 0.0012}$	$\mathbf{0.000114 \pm 0.00017}$
	Heteroscedastic LVGP	0.28 ± 1	0.238 ± 0.59	0.0426 ± 0.081	0.0488 ± 0.15
	Separate GP	0.709 ± 1.3	0.238 ± 0.59	0.0349 ± 0.062	0.0684 ± 0.19
	Categorical kernel	$\mathbf{0.00491 \pm 0.006}$	0.0548 ± 0.066	0.0025 ± 0.011	0.000646 ± 0.0023
EI	Standard LVGP	0.0102 ± 0.0098	$\mathbf{0.00181 \pm 0.0076}$	$\mathbf{-6.54{\times}10^{-7} \pm 4.2{\times}10^{-7}}$	$\mathbf{1.81{\times}10^{-6} \pm 5.8{\times}10^{-6}}$
	Heteroscedastic LVGP	0.022 ± 0.031	0.0802 ± 0.058	0.0225 ± 0.048	0.0324 ± 0.12
	Separate GP	0.422 ± 1	0.662 ± 0.95	0.029 ± 0.049	0.0457 ± 0.18
	Categorical kernel	$\mathbf{0.01 \pm 0.0092}$	0.158 ± 0.39	0.000174 ± 0.00025	0.000521 ± 0.00094
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0259 ± 0.023	$\mathbf{0.0211 \pm 0.035}$	$\mathbf{0.00868 \pm 0.021}$	0.0163 ± 0.087
	Separate GP	0.425 ± 1	0.434 ± 0.74	0.0288 ± 0.049	0.0486 ± 0.15
	Categorical kernel	$\mathbf{0.00801 \pm 0.0085}$	0.0421 ± 0.048	0.0095 ± 0.021	$\mathbf{0.000346 \pm 0.00085}$
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.0208 ± 0.019	$\mathbf{0.00541 \pm 0.014}$	$\mathbf{0.0194 \pm 0.028}$	$\mathbf{6.35{\times}10^{-5} \pm 0.00015}$
	Separate GP	0.116 ± 0.55	0.666 ± 1.1	0.0413 ± 0.048	0.000237 ± 0.00041
	Categorical kernel	$\mathbf{0.00552 \pm 0.0094}$	0.135 ± 0.41	0.0406 ± 0.049	0.000346 ± 0.00089
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$\mathbf{0.0203 \pm 0.025}$	$\mathbf{0.00189 \pm 0.0049}$	$\mathbf{0.0085 \pm 0.021}$	0.0162 ± 0.087
	Separate GP	0.723 ± 1.3	0.288 ± 0.78	0.0768 ± 0.094	0.168 ± 0.3
	Categorical kernel	0.397 ± 1	0.154 ± 0.57	0.0411 ± 0.1	$\mathbf{0.000219 \pm 0.00055}$

Table 3: Final noise σ^2 at the incumbent design for every acquisition × surrogate model ($n_{\text{rep}} = 3$, mean ± sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	$\mathbf{0.01 \pm 5.3{\times}10^{-8}}$	23.8 ± 22	0.00608 ± 0.0012	0.00731 ± 0.0062
	Heteroscedastic LVGP	$0.01 \pm 9{\times}10^{-7}$	26.6 ± 22	$\mathbf{0.00576 \pm 0.0016}$	0.00731 ± 0.0062
	Separate GP	0.016 ± 0.012	23.7 ± 27	0.00892 ± 0.0073	0.0122 ± 0.015
	Categorical kernel	0.0111 ± 0.0054	$\mathbf{10.8 \pm 19}$	0.00608 ± 0.0012	$\mathbf{0.00647 \pm 0.0045}$
PI	Standard LVGP	$0.01 \pm 2.2{\times}10^{-7}$	$\mathbf{12 \pm 19}$	$0.0064 \pm 2.3{\times}10^{-6}$	$\mathbf{0.00564 \pm 1.8{\times}10^{-6}}$
	Heteroscedastic LVGP	0.0118 ± 0.0059	24.7 ± 24	0.0066 ± 0.0048	0.00731 ± 0.0062
	Separate GP	0.011 ± 0.0054	12.9 ± 19	$\mathbf{0.00556 \pm 0.0034}$	0.00663 ± 0.0045
	Categorical kernel	$\mathbf{0.01 \pm 1.7{\times}10^{-7}}$	15.9 ± 21	$0.0064 \pm 3.9{\times}10^{-6}$	$0.00564 \pm 5.3{\times}10^{-6}$
EI	Standard LVGP	$\mathbf{0.01 \pm 1.5{\times}10^{-7}}$	$\mathbf{10.4 \pm 18}$	$0.0064 \pm 7.5{\times}10^{-9}$	$0.00564 \pm 4.8{\times}10^{-7}$
	Heteroscedastic LVGP	$0.01 \pm 5.4{\times}10^{-7}$	23.4 ± 22	0.00592 ± 0.0014	0.00731 ± 0.0062
	Separate GP	0.012 ± 0.0075	14.2 ± 18	$\mathbf{0.00524 \pm 0.0035}$	0.00731 ± 0.0062
	Categorical kernel	$0.01 \pm 1.8{\times}10^{-7}$	17.3 ± 21	$0.0064 \pm 1.2{\times}10^{-6}$	$\mathbf{0.00564 \pm 3.5{\times}10^{-6}}$
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	$0.01 \pm 2.7{\times}10^{-7}$	25 ± 22	0.00608 ± 0.0012	$\mathbf{0.00564 \pm 4.8{\times}10^{-6}}$
	Separate GP	0.012 ± 0.0075	16.4 ± 21	$\mathbf{0.0054 \pm 0.0034}$	0.00662 ± 0.0045
	Categorical kernel	$\mathbf{0.01 \pm 1.5{\times}10^{-7}}$	$\mathbf{14.9 \pm 21}$	$0.0064 \pm 3.1{\times}10^{-6}$	$0.00564 \pm 5.1{\times}10^{-6}$
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	$0.01 \pm 4.8{\times}10^{-7}$	15.1 ± 21	0.00544 ± 0.0019	$\mathbf{0.00564 \pm 2.3{\times}10^{-6}}$
	Separate GP	$0.01 \pm 6.2{\times}10^{-7}$	13.6 ± 21	0.00505 ± 0.0045	0.00731 ± 0.0062
	Categorical kernel	$\mathbf{0.01 \pm 6{\times}10^{-7}}$	$\mathbf{9.12 \pm 17}$	$\mathbf{0.00465 \pm 0.0023}$	$0.00564 \pm 7.5{\times}10^{-6}$
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$\mathbf{0.01 \pm 5.1{\times}10^{-7}}$	$\mathbf{1.41 \pm 0.0085}$	$\mathbf{0.00576 \pm 0.0016}$	0.00647 ± 0.0045
	Separate GP	0.017 ± 0.013	7.48 ± 18	0.00839 ± 0.0071	0.0131 ± 0.016
	Categorical kernel	0.011 ± 0.0054	4.5 ± 12	0.00608 ± 0.0012	$\mathbf{0.00564 \pm 5{\times}10^{-6}}$

Table 4: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	$\mathbf{0.01 \pm 2.6{\times}10^{-8}}$	$\mathbf{1.41 \pm 0.0058}$	0.00608 ± 0.0012	0.00731 ± 0.0062
	Heteroscedastic LVGP	$0.01 \pm 4.1{\times}10^{-7}$	27.9 ± 22	$\mathbf{0.0056 \pm 0.0018}$	0.00731 ± 0.0062
	Separate GP	0.017 ± 0.013	22.8 ± 24	0.00788 ± 0.0068	0.0121 ± 0.013
	Categorical kernel	0.0121 ± 0.0075	20.2 ± 22	0.00712 ± 0.0034	$\mathbf{0.00648 \pm 0.0045}$
PI	Standard LVGP	$0.01 \pm 1.9{\times}10^{-7}$	$\mathbf{2.89 \pm 8}$	$0.0064 \pm 2.5{\times}10^{-6}$	$0.00564 \pm 1.7{\times}10^{-6}$
	Heteroscedastic LVGP	0.0115 ± 0.0059	16 ± 22	0.00736 ± 0.0051	0.00814 ± 0.0075
	Separate GP	0.0161 ± 0.012	21.7 ± 24	$\mathbf{0.00584 \pm 0.0043}$	0.00746 ± 0.0063
	Categorical kernel	$\mathbf{0.01 \pm 4.3{\times}10^{-8}}$	11.4 ± 18	0.00624 ± 0.00086	$\mathbf{0.00564 \pm 3.3{\times}10^{-6}}$
EI	Standard LVGP	$0.01 \pm 1.3{\times}10^{-7}$	$\mathbf{1.41 \pm 0.0072}$	$0.0064 \pm 8.6{\times}10^{-10}$	$0.00564 \pm 2.3{\times}10^{-7}$
	Heteroscedastic LVGP	$0.01 \pm 9.3{\times}10^{-7}$	20.9 ± 22	0.00572 ± 0.0033	0.00731 ± 0.0062
	Separate GP	0.014 ± 0.01	25 ± 26	$\mathbf{0.00524 \pm 0.0035}$	0.00662 ± 0.0045
	Categorical kernel	$\mathbf{0.01 \pm 9.3{\times}10^{-8}}$	16.8 ± 21	$0.0064 \pm 5.1{\times}10^{-7}$	$\mathbf{0.00564 \pm 3.5{\times}10^{-6}}$
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	$0.01 \pm 4.7{\times}10^{-7}$	$\mathbf{2.83 \pm 7.7}$	0.00576 ± 0.0016	0.00647 ± 0.0045
	Separate GP	0.014 ± 0.01	21.8 ± 23	$\mathbf{0.00524 \pm 0.0035}$	0.00814 ± 0.0075
	Categorical kernel	$\mathbf{0.01 \pm 7.3{\times}10^{-8}}$	7.1 ± 15	0.00576 ± 0.0016	$\mathbf{0.00564 \pm 2.8{\times}10^{-6}}$
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	$0.01 \pm 3.2{\times}10^{-7}$	$\mathbf{1.41 \pm 0.013}$	0.00496 ± 0.0022	$0.00564 \pm 1.3{\times}10^{-6}$
	Separate GP	0.011 ± 0.0054	12.1 ± 22	$\mathbf{0.00428 \pm 0.0037}$	$\mathbf{0.00564 \pm 2.5{\times}10^{-6}}$
	Categorical kernel	$\mathbf{0.01 \pm 9{\times}10^{-8}}$	4.64 ± 9.8	0.00445 ± 0.0037	$0.00564 \pm 2.7{\times}10^{-6}$
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$\mathbf{0.01 \pm 5.5{\times}10^{-7}}$	$\mathbf{1.41 \pm 0.0073}$	0.00576 ± 0.0016	0.00647 ± 0.0045
	Separate GP	0.017 ± 0.013	3.41 ± 9.6	0.00788 ± 0.0068	0.0102 ± 0.0092
	Categorical kernel	0.0131 ± 0.009	2.34 ± 4.7	$\mathbf{0.00563 \pm 0.0033}$	$\mathbf{0.00564 \pm 2.5{\times}10^{-6}}$

Table 5: Final mean-variance regret (RAHBO robustness, MV $= f + \alpha\sigma^2$, $\alpha = 1$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	$\mathbf{0.00641 \pm 0.0063}$	$\mathbf{0.344 \pm 0.54}$	$\mathbf{0.00381 \pm 0.014}$	0.0339 ± 0.13
	Heteroscedastic LVGP	0.0302 ± 0.035	0.793 ± 1.1	0.00794 ± 0.019	0.0343 ± 0.13
	Separate GP	0.63 ± 1.2	2.09 ± 3.3	0.0937 ± 0.11	0.178 ± 0.34
	Categorical kernel	0.197 ± 0.71	1.11 ± 3.9	0.00409 ± 0.014	$\mathbf{0.0176 \pm 0.091}$
PI	Standard LVGP	$\mathbf{0.0127 \pm 0.015}$	0.571 ± 1.4	$\mathbf{0.000738 \pm 0.0011}$	$\mathbf{0.000135 \pm 0.00025}$
	Heteroscedastic LVGP	0.549 ± 1.2	2.53 ± 4.5	0.0397 ± 0.075	0.0362 ± 0.13
	Separate GP	0.121 ± 0.56	0.736 ± 1.5	0.0251 ± 0.049	0.061 ± 0.19
	Categorical kernel	0.0145 ± 0.012	$\mathbf{0.273 \pm 0.46}$	0.000797 ± 0.0019	0.00107 ± 0.0013
EI	Standard LVGP	$\mathbf{0.0123 \pm 0.012}$	$\mathbf{0.0884 \pm 0.13}$	$\mathbf{3.12{\times}10^{-7} \pm 3.7{\times}10^{-6}}$	$\mathbf{7.81{\times}10^{-6} \pm 3.4{\times}10^{-5}}$
	Heteroscedastic LVGP	0.0212 ± 0.024	0.516 ± 1	0.00649 ± 0.017	0.0386 ± 0.13
	Separate GP	0.223 ± 0.77	1.19 ± 1.7	0.0281 ± 0.05	0.0343 ± 0.13
	Categorical kernel	0.0132 ± 0.014	0.332 ± 0.54	0.000436 ± 0.00059	0.00044 ± 0.00073
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0166 ± 0.016	0.412 ± 0.78	0.00432 ± 0.014	$\mathbf{0.000791 \pm 0.0013}$
	Separate GP	0.219 ± 0.78	0.527 ± 1.1	0.0261 ± 0.05	0.0471 ± 0.18
	Categorical kernel	$\mathbf{0.0122 \pm 0.012}$	$\mathbf{0.294 \pm 0.54}$	$\mathbf{0.000882 \pm 0.0015}$	0.000907 ± 0.0011
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.024 ± 0.025	$\mathbf{0.424 \pm 0.88}$	$\mathbf{0.0127 \pm 0.022}$	$\mathbf{0.000224 \pm 0.00067}$
	Separate GP	0.027 ± 0.027	0.523 ± 1.1	0.0453 ± 0.063	0.0357 ± 0.13
	Categorical kernel	$\mathbf{0.0191 \pm 0.024}$	0.474 ± 1.1	0.025 ± 0.026	0.00284 ± 0.0053
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$\mathbf{0.0264 \pm 0.024}$	$\mathbf{0.00144 \pm 0.0029}$	0.00789 ± 0.019	0.0178 ± 0.091
	Separate GP	0.73 ± 1.3	0.915 ± 2.1	0.102 ± 0.13	0.195 ± 0.34
	Categorical kernel	0.111 ± 0.56	0.288 ± 1.1	$\mathbf{0.00431 \pm 0.014}$	$\mathbf{0.000752 \pm 0.002}$

Table 6: Final mean-variance regret (RAHBO robustness, MV $= f + \alpha\sigma^2$, $\alpha = 1$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	**0.00408 ± 0.0047**	**0.00407 ± 0.0032**	**0.00381 ± 0.014**	0.0339 ± 0.13
	Heteroscedastic LVGP	0.0188 ± 0.021	1 ± 1.6	0.0097 ± 0.021	0.034 ± 0.13
	Separate GP	0.733 ± 1.3	2.34 ± 3.4	0.0784 ± 0.1	0.207 ± 0.34
	Categorical kernel	0.297 ± 0.88	1.77 ± 4.3	0.019 ± 0.064	**0.0193 ± 0.1**
PI	Standard LVGP	0.0131 ± 0.014	**0.139 ± 0.74**	**0.000902 ± 0.0012**	**0.000115 ± 0.00017**
	Heteroscedastic LVGP	0.281 ± 1	2.41 ± 4.5	0.0436 ± 0.085	0.0513 ± 0.15
	Separate GP	0.715 ± 1.3	1.72 ± 2.2	0.0343 ± 0.064	0.0702 ± 0.2
	Categorical kernel	**0.00491 ± 0.006**	0.854 ± 3.3	0.00234 ± 0.01	0.000646 ± 0.0023
EI	Standard LVGP	0.0102 ± 0.0098	**0.00449 ± 0.0035**	**$-6.55\times10^{-7} \pm 4.2\times10^{-7}$**	**$1.67\times10^{-6} \pm 5.6\times10^{-6}$**
	Heteroscedastic LVGP	0.022 ± 0.031	0.868 ± 1.7	0.0218 ± 0.049	0.0341 ± 0.13
	Separate GP	0.426 ± 1.1	1.81 ± 2	0.0278 ± 0.05	0.0467 ± 0.18
	Categorical kernel	**0.01 ± 0.0092**	0.765 ± 1.3	0.000174 ± 0.00025	0.000521 ± 0.00094
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0259 ± 0.023	0.17 ± 0.79	**0.00804 ± 0.019**	0.0171 ± 0.091
	Separate GP	0.429 ± 1.1	1.29 ± 1.8	0.0277 ± 0.05	0.0511 ± 0.15
	Categorical kernel	**0.00801 ± 0.0085**	**0.0895 ± 0.19**	0.00886 ± 0.02	**0.000347 ± 0.00085**
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.0208 ± 0.019	**0.00777 ± 0.022**	**0.018 ± 0.026**	**$6.35\times10^{-5} \pm 0.00015$**
	Separate GP	0.117 ± 0.56	1.42 ± 2.1	0.0392 ± 0.049	0.000237 ± 0.00042
	Categorical kernel	**0.00552 ± 0.0094**	0.159 ± 0.47	0.0387 ± 0.05	0.000346 ± 0.00088
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	**0.0203 ± 0.025**	**0.00399 ± 0.012**	**0.00786 ± 0.019**	0.017 ± 0.091
	Separate GP	0.73 ± 1.3	0.581 ± 1.5	0.0783 ± 0.1	0.173 ± 0.3
	Categorical kernel	0.4 ± 1	0.717 ± 3.1	0.0403 ± 0.1	**0.00022 ± 0.00055**